



PsyMatch



AI-Driven Framework



Retrospective Data Analysis

● CLINICAL DECISION SUPPORT PROTOTYPE

Personalizing Mental Health Treatment with AI

An educational framework leveraging machine learning, causal inference, and pattern mining to address variability in patient outcomes.

PROJECT LEAD



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Data Science & AI Research

FOCUS AREA

Precision Medicine

METHODOLOGY

Causal ML & Mining

DATE

2026

Why Personalized Treatment Is Hard

Outcomes vary drastically across patients. The "trial-and-error" approach delays recovery and increases distress. Why is it so complex?



Symptom Severity

Differences in symptom intensity require precisely calibrated intervention strengths. What works for mild cases fails for severe ones.



Demographics

Age, gender, and cultural background influence both physiological response to medication and social support needs.



Lifestyle Factors

Sleep quality, daily activity, and stress levels act as crucial moderators, significantly impacting treatment success rates.



Treatment Mix

Complex interactions between various medications and therapy types complicate prediction. Mono-therapy vs. combinations yield different results.



Our Motivation

There is a critical need for **AI decision support** trained on historical data to replace guesswork with **evidence-based precision**.



Data-Driven



Precision Care

Project Objective

To address treatment variability, we aim to build an educational clinical decision support prototype.

 Advisory tool; not a diagnostic system



01

Build System

Develop a clinical decision support framework to assist practitioners in selecting optimal treatments based on patient profiles.



02

Analyze Patterns

Mine historical treatment data to identify effective strategies and successful outcome patterns across diverse patient cohorts.



03

Estimate Impact

Apply propensity score matching to control for confounders and estimate the true causal effect of treatments, ensuring unbiased results.



04

Rank Outcomes

Predict individualized outcomes and generate a ranked list of treatment recommendations tailored specifically for each patient profile.

Dataset Overview

A structured, retrospective dataset forms the backbone of our analysis, capturing rich clinical details while ensuring privacy.



**DATASET TYPE**
Retrospective Analysis

**CLINICAL SCALE**
Self-Reported 1-10

**DATA PRIVACY**
Anonymized Records

Features: Patient & Clinical

Demographics

Age: Normalized numeric value representing patient age.

Gender: Categorical variable for biological sex or gender identity.

Clinical Profile

Diagnosis: Text label description of the condition.

Symptom Severity: Self-reported score (1-10).

Mood Score: Baseline mood rating (1-10).

Lifestyle Metrics

Sleep Quality: Self-reported quality rating (1-10).

Activity Level: Physical activity in hours/week.

Stress Level: Perceived stress score (1-10).

AI-Derived Signals

Emotional State: AI-labeled emotional state provided as a categorical feature in the dataset.

Features: Treatment & Outcome

We map detailed intervention variables to concrete success metrics, focusing on a clear probabilistic target for improvement.

➔ INTERVENTION VARIABLES (INPUTS)



Medication

Specific drug class prescribed

Categorical



Therapy Type

CBT, Talk Therapy, etc.

Categorical



Duration

Treatment length in weeks

Numeric



Adherence

Patient compliance percentage

%



🚩 SUCCESS METRICS (TARGETS)



Outcome Status

Improved, No Change, Deteriorated

Class



Progress Score

Score 1-10 indicating progress

Numeric

🎯 PRIMARY PREDICTION TARGET



Probability of Improvement

Model-ready binary or probabilistic indicator derived from outcome measures.

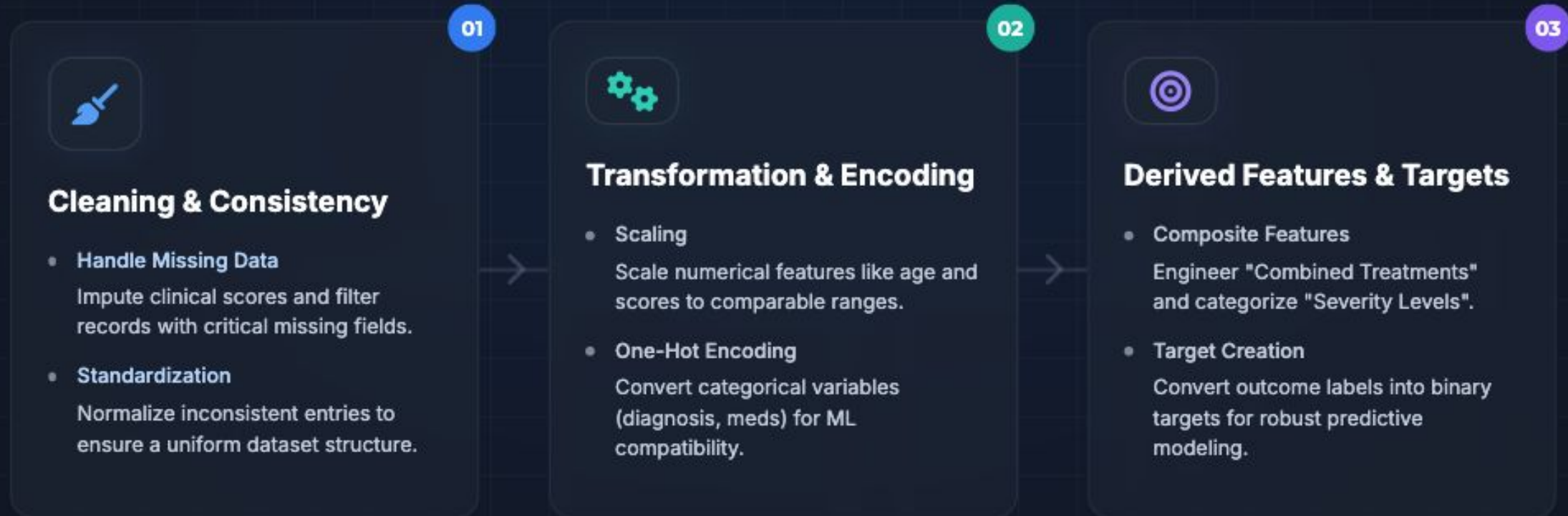
An End-to-End Pipeline

From raw retrospective data to actionable clinical recommendations, our system follows a rigorous nine-step workflow.



Preprocessing & Feature Engineering

Before modeling, we rigorously prepare the data by addressing inconsistencies, transforming variables, and engineering predictive signals.



Treatment Effectiveness

Before personalization, we analyze aggregate outcomes to identify effective strategies across the population.

Improvement Rate by Strategy

Metric: % Patients Improved



Simulated data showing combined therapies often yield higher improvement rates.

Severity vs. Outcome Distribution

Count: Patient Cohort



Association Rule Mining

Discovering "If-Then" Patterns

We utilize the Apriori algorithm to surface frequent relationships between patient characteristics and effective treatments, providing interpretable clinical insights.

IF [Diagnosis = Anxiety] & [Stress = High]
THEN [Best Therapy = CBT]

Evaluation Metrics

Support

How frequently the rule appears in the dataset.

Confidence

How often the rule has been found to be true.

Lift

Strength of association vs. random chance.

Propensity Score Matching

To isolate the true impact of treatment, we must move beyond simple correlation. We transform observational data into balanced cohorts, mimicking a randomized trial.



Average Treatment Effect (ATE)

The isolated impact of the intervention on the matched population.

$$ATE = E[Y(1) - Y(0)]$$

Learning Outcome Patterns



EVALUATION FOCUS



PRECISION

Minimizing false positives to avoid giving patients false hope.



RECALL

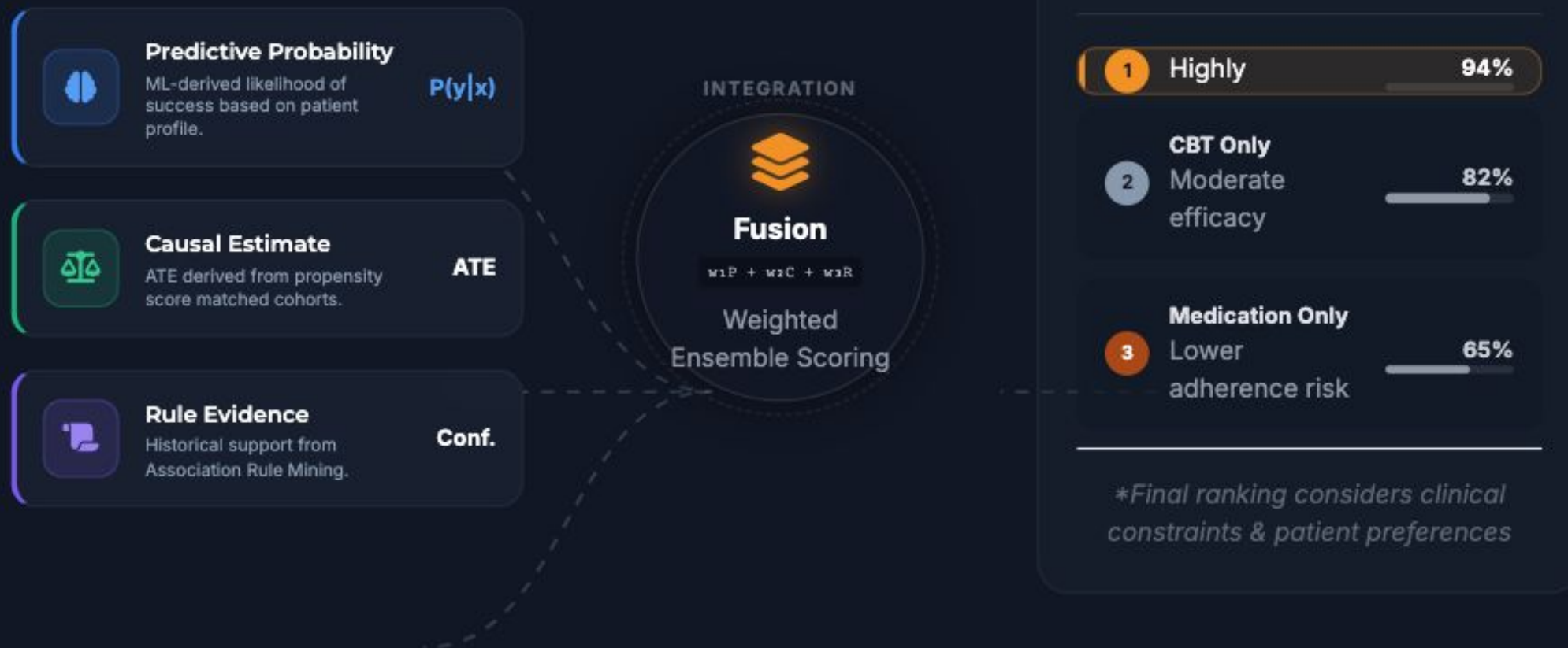
Ensuring potential improvement opportunities aren't missed.



INTERNAL VALIDATION

Models evaluated for plausibility; not clinically calibrated.

Multi-Signal Fusion Strategy



Interface & Output

The project includes an interactive, notebook-based interface built with ipywidgets that allows users to input patient characteristics and view generated treatment recommendations within the Jupyter environment.

Patient input (left panel)

- Diagnosis (dropdown)
- Age (slider), Gender (radio)
- Symptom Severity (1–10 slider)
- Mood Score (1–10 slider)
- Sleep Quality (1–10 slider)
- Physical Activity (hours/week slider)
- Stress Level (1–10 slider)
- AI Emotional State (dropdown)
- **Button: "Get Treatment Recommendations"**

Output (right panel)

- Patient profile summary (echo inputs)
- Top-N recommended treatment combinations
- Model-derived likelihood indicators (High/Medium/Low)
- Number of similar patients referenced
- Historical improvement rate
- Brief rule-based insight text

Runs entirely inside Jupyter/Colab using ipywidgets; button triggers a backend Python function; no persistence, accounts, EHR integration, or deployment—demonstration and analysis only.

Ethics & Limitations



Advisory Tool - NOT a Diagnostic System

PsyMatch is designed as an educational prototype to support clinical decision-making. It does not replace professional medical judgment, diagnosis, or treatment planning.

! KEY LIMITATIONS



Retrospective Bias

Models trained on historical data may inadvertently learn and perpetuate past prescribing habits rather than optimal care standards, reflecting "what was done" rather than "what works best."



Data Scope & Completeness

The current dataset does not fully account for rare comorbidities, specific drug-drug interactions, or genetic factors, creating blind spots for complex patient profiles.

🛡️ ETHICAL PRINCIPLES



Privacy & Anonymity

Strict adherence to data de-identification protocols. The dataset is used solely for educational and research purposes and contains no real patient identifiers (PHI).



Human Oversight

Algorithm-in-the-loop approach ensures that final treatment decisions always remain with the clinician. The AI serves as a "second opinion," not a decision-maker.

Conclusion & Future Work

Achievements

We developed a comprehensive framework integrating Machine Learning, Causal Inference, and Pattern Mining. This project demonstrates the feasibility of using historical data to provide personalized, evidence-based treatment support via a prototype.



Fairness Analysis

Audit for algorithmic bias across demographics.



External Validation

Test models on diverse, multi-center datasets.



Deployment

Pilot real-time integration into EHR systems. Future UI work: consider migrating to Streamlit/Flask for a deployable prototype; current interface is notebook-based for demonstration.